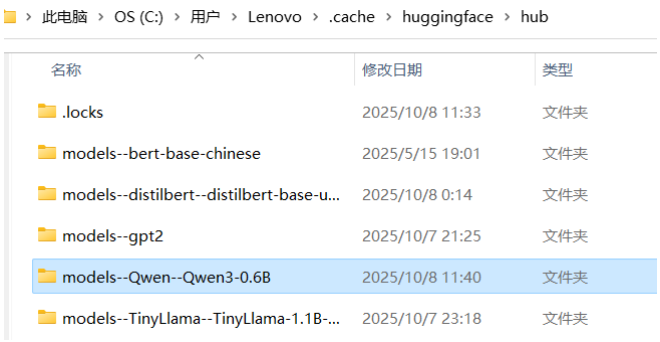
1 作业内容

①下载并安装PyTorch和Transformers
②使用Transformers库实现一个模型(任意)进行任意数量轮次的对话(最好超过两轮)，并保留历史记录;
③熟悉分词，并分别将一个句子(任何)转换为其分词形式和ID形式。
④熟悉模型的结构，并打印该模型任意层的一个(或多个)attention层的输出。

2 代码实现

2.1 多轮对话

目前下载好的PyTorch版本为2.5.1+cu121，Transformers版本4.52.3，CUDA版本 12.1 (nvcc 12.1.66)



选择用Qwen/Qwen3-0.6B模型，轻量且效果好。

```
from transformers import AutoTokenizer, AutoModelForCausalLM
model_name="Qwen/Qwen3-0.6B"
#导入模型
tokenizer=AutoTokenizer.from_pretrained(model_name)
model=AutoModelForCausalLM.from_pretrained(model_name)

chat_history = []#全局变量记录历史记录
```

如果要实现带历史记录的输入，可以用tokenizer.apply_chat_template方法，可以将对话数据格式化为可接受的输入格式，支持Qwen的对话模板，能够自动处理历史记录。

chat_history以**列表嵌套字典**的格式存储历史记录，tokenizer.apply_chat_template可以自动将其转化为对话格式。

定义一个函数来调用模型进行对话。函数输入为User新的语句，模型输入为整个的历史记录，输出为模型的最后一次输出。

```
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch

model_name="Qwen/Qwen3-0.6B"

tokenizer=AutoTokenizer.from_pretrained(model_name)
model=AutoModelForCausalLM.from_pretrained(model_name)

chat_history=[]

def chat(user_input,max_new_tokens=256):
    global chat_history
```

```
    chat_history.append({"role": "user", "content": user_input})
    templated_input=tokenizer.apply_chat_template(
        chat_history,
        tokenize=False,
        add_generation_prompt=True
    )
    #分词转为张量
    inputs=tokenizer(templated_input, return_tensors="pt").to(model.device)
    outputs=model.generate(
        **inputs,
        max_new_tokens=max_new_tokens,
    )
    #解码输出
    reply=tokenizer.decode(outputs[0], skip_special_tokens=True)
    #提取Assistant的新回复部分
    response=reply.split("assistant")[-1].replace(":", "").strip()
    #存入历史记录
    chat_history.append({"role": "assistant", "content": response})
    return response
```

输入对话：

```
print("User: Hello, who are you?")
print("Assistant:", chat("Hello, who are you?"))

print("\nUser:What is the capital of China?")
print("Assistant:", chat("What is the capital of China?"))

print("\nUser:What local specialties are there here?")
print("Assistant:", chat("What local specialties are there here?"))

print("\nUser:What famous attractions are there here?")
print("Assistant:", chat("What famous attractions are there here?"))
```

输出结果如下，Qwen的输出格式带有思考过程，即<think>，同时也测试了其他问题，偶尔会出现客观上的错误，可能是模型小的原因，没有那么大的数据量来训练。

```
User: Hello, who are you?
Assistant: designed to help with questions and tasks. How can I assist you today? 😊

User:What is the capital of China?
Assistant: <think>
Okay, the user asked, "What is the capital of China?" I need to recall the correct answer. China's capital is Beijing. I should confirm that there's no
confusion with other countries' capitals. Let me make sure there's no other city that's commonly known as the capital. Beijing is the only one I can thi
nk of. So, the response should be straightforward and accurate.
</think>

The capital of China is **Beijing**.

User:What local specialties are there here?
Assistant: <think>
Okay, the user just asked about local specialties in China. Let me think. I need to provide a helpful response. China has many cities and regions with u
nique cuisines. The user might be looking for food options that are popular in their area. Since they asked for local specialties, I should mention the
main cities and their famous dishes.

First, I should list some major cities in China and their well-known dishes. Beijing is a big city, so mentioning Beijing's famous dishes like duck egg
and dumplings would be good. Then, I can include other cities like Shanghai, where seafood is a big deal. It's important to note that the user might be
in a city and wants to know about local food options, so providing a variety of cities and their specialties would be helpful.

I should also mention that the user might want to know more about specific dishes or regions. Including a friendly note at the end to ask if they need m
ore information would be a good way to conclude the response. Need to make sure the information is accurate and helpful.
</think>

China has a rich culinary heritage with many local specialties across cities. Here are some popular ones
- **Beijing** Famous dishes like **duck egg and dumplings**, **soufu**, and
```

```
User:What famous attractions are there here?
Assistant: <think>
Okay, the user asked about famous attractions in China. Let me start by recalling the major cities and their famous landmarks. Beijing is a good start w
ith the Summer Palace and the Great Wall. Then there's Shanghai with the Bund and the Shanghai Tower.

Next, I should mention other major cities like Guangzhou, Shenzhen, and Harbin. Each city has unique attractions. Guangzhou is known for its famous mark
ets and the Golden Gate Bridge. Shenzhen is a tech hub with the Shanghai Tower and the Hongqiao Lake. Harbin has the Olympic Park and the Great Northern
Mountains.

I need to make sure the answer is comprehensive but not too long. Also, check if the user wants more details on each attraction. Maybe add a note about
cultural significance to give context. Avoid any errors in the names and information. Keep the tone friendly and helpful.
</think>

China has many iconic attractions across major cities
- **Beijing** Summer Palace (Yuanmingyuan), Great Wall, and the Forbidden City.
- **Shanghai** Bund, Shanghai Tower, and Hongqiao Lake.
- **Guangzhou** Golden Gate Bridge and the famous Guangdong Market.
- **Shenzhen** Shanghai Tower, Hongqiao Lake
```

2.2 分词

每个模型自带一个分词器tokenizer，可以将一句话分为各个带有id的tokens。以便进行后续的训练。

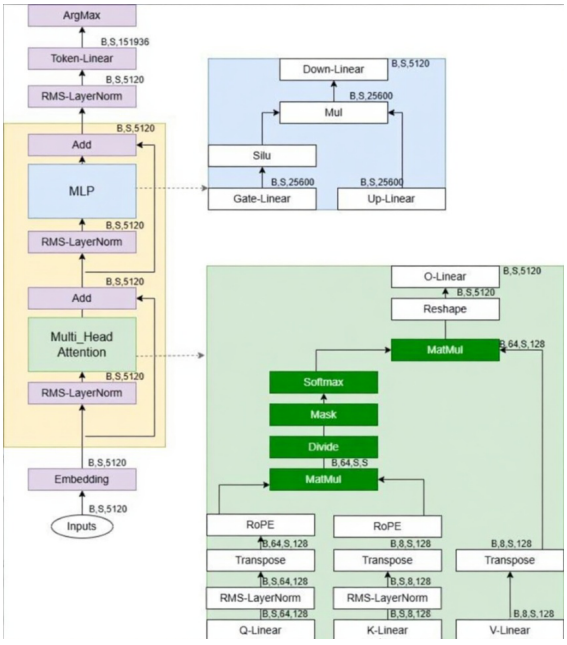
```
sentence="I like the class named GenAI,though i don't use transformers before."
tokens=tokenizer.tokenize(sentence)
token_ids=tokenizer.convert_tokens_to_ids(tokens)

print("Tokens:", tokens)
print("Token IDs:", token_ids)
```

```
Tokens: ['I', 'Ġlike', 'Ġthe', 'Ġclass', 'Ġnamed', 'ĠGen', 'AI', ',', 'though', 'Ġi', 'Ġdon', "'t", 'Ġuse', 'Ġtransformers', 'Ġbefore', '.']
Token IDs: [40, 1075, 279, 536, 6941, 9316, 15469, 11, 4535, 600, 1513, 944, 990, 86870, 1573, 13]
```

2.3 Attention层输出

图为Qwen3-32B模型的结构，0.6B只是参数更小：



据Qwen3的模型文档以及代码，Qwen3共有28层，每层共有16个注意力头。

可在outputs时设置是否输出attention层，此处输出第0层的attention层的16个attention头的输出

```
inputs=tokenizer("What is the capital of China?", return_tensors="pt")
outputs=model(**inputs, output_attentions=True)

print(len(outputs.attentions))
print(outputs.attentions[0].shape) #(batch,heads,seq_len,seq_len)
print(outputs.attentions[0])
```

```
28
torch.Size([1, 16, 7, 7])
```

```
tensor([[[[1.0000e+00, 0.0000e+00, 0.0000e+00, 0.0000e+00, 0.0000e+00,
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           6.9829e-06, 2.3529e-01]],

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          [3.5965e-06, 1.3277e-05, 9.0734e-06, 9.9997e-01, 0.0000e+00,
           0.0000e+00, 0.0000e+00],
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           9.9655e-01, 0.0000e+00],
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